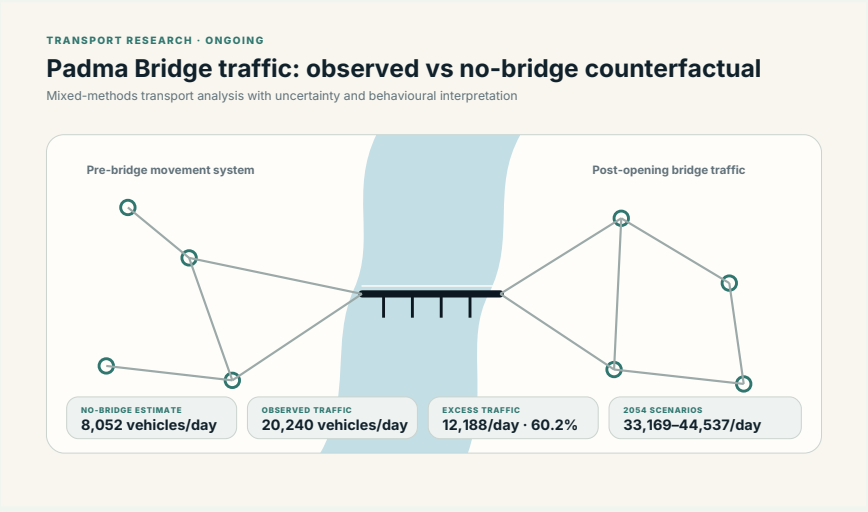
SELECTED ACADEMIC WORKS / 2026

TRANSPORTATION & URBAN SYSTEMS / CLIMATE RESILIENCE / GEOSPATIAL ANALYSIS

# Rifat Hossain Talukder

I study how major infrastructure and environmental change reshape mobility, urban growth and access. My work combines GIS, Earth observation, spatial modelling, mixed methods and geospatial machine learning to answer planning questions.




Padma Bridge traffic counterfactual framework

Khulna University · Bangladesh

rifatht21@gmail.com

OVERVIEW

# Research profile

Planning research grounded in spatial evidence, quantitative analysis and field or archival interpretation.

## Research focus

How infrastructure and environmental change alter mobility, settlement patterns, physical exposure and access to essential services.

## Working approach

Spatial data identify change; quantitative and geospatial models test possible relationships; qualitative and archival evidence support interpretation.

THREE INTERCONNECTED THEMES

- **Transport, mobility & infrastructure**
- **Climate risk & resilient access**
- **Geospatial evidence & modelling**

AREAS OF PRACTICE

### Transport

Counterfactual traffic analysis, route choice, travel behaviour and network accessibility.

### Environment

Satellite-derived indices, flood vulnerability, land cover and exposure.

### Methods

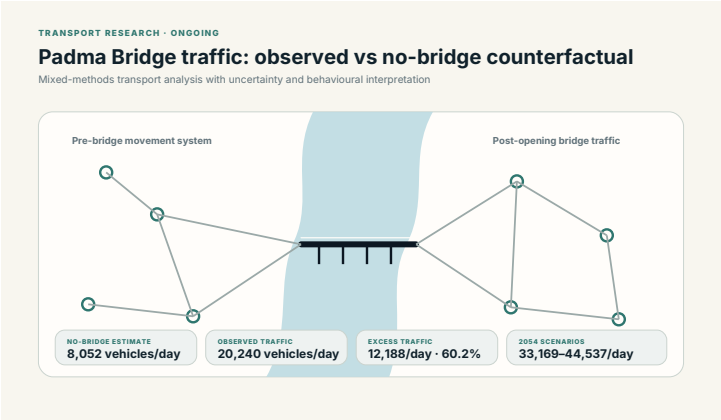
GIS, Random Forest, scenario modelling, uncertainty and mixed methods.

RESEARCH 01 · ONGOING

# Traffic after the Padma Bridge

In this study, I estimate a present-day no-bridge counterfactual, compare it with observed bridge traffic, examine behavioural change and test long-term traffic scenarios under uncertainty.

| OBSERVED BRIDGE TRAFFIC | PREFERRED NO-BRIDGE ESTIMATE | BRIDGE-ATTRIBUTABLE EXCESS |
|---|---|---|
| 20,240 | 8,052 | 12,188 |
| vehicles per day | vehicles per day | vehicles per day |




Conceptual comparison of observed bridge traffic with a modelled no-bridge condition.

## Research problem

The Padma Bridge replaced a ferry- and launch-dependent crossing system with a fixed road connection. To estimate its effect on traffic, I compare observed bridge crossings with a modelled no-bridge scenario, accounting for background growth, route diversion, mode transfer and previously suppressed demand.

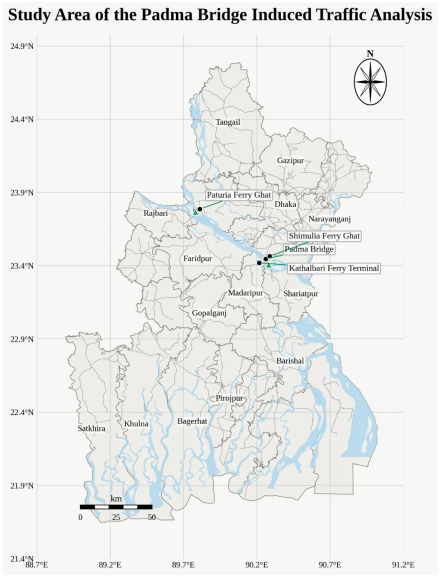
## Mixed-methods design

I combine temporal and spatial counterfactual reasoning. Pre-bridge movement trends provide the main temporal no-bridge estimate, while district-level built-up area and census information support spatial interpretation and sensitivity checks. I use Monte Carlo simulation to represent uncertainty instead of treating the counterfactual as a single known value. Interviews help me distinguish reported additional travel from destination change, mode transfer and route diversion.

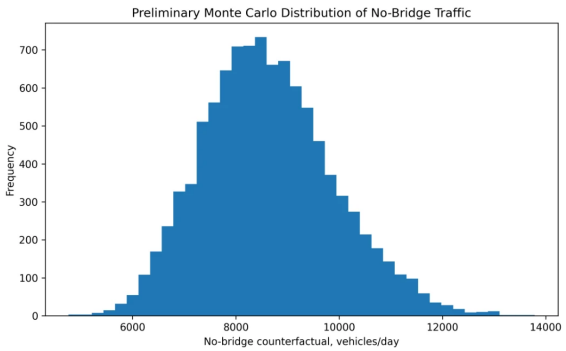
RESEARCH 01 · EVIDENCE

# Mapping the traffic mechanisms

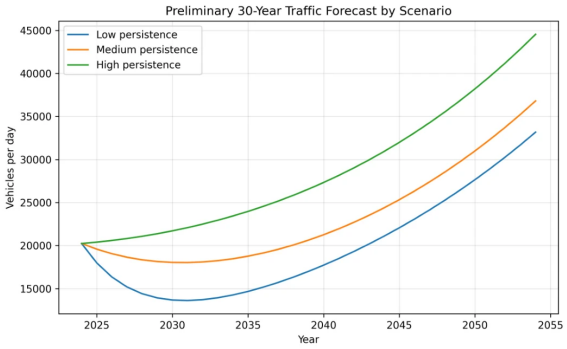
Study-area mapping, the no-bridge traffic estimate and long-term demand scenarios.




Study-area mapping from the working draft.




Monte Carlo distribution of simulated no-bridge traffic.




Scenario-based 2054 demand range: 33,169–44,537 vehicles/day.

## Behavioural interpretation

Interviews identified increased trip frequency, newly feasible journeys, destination changes, mode transfer and route diversion. These responses help distinguish additional travel from shifts in existing movement.

RESEARCH 02 · COMPLETED UNDP WORK

# Cyclone-induced flood vulnerability

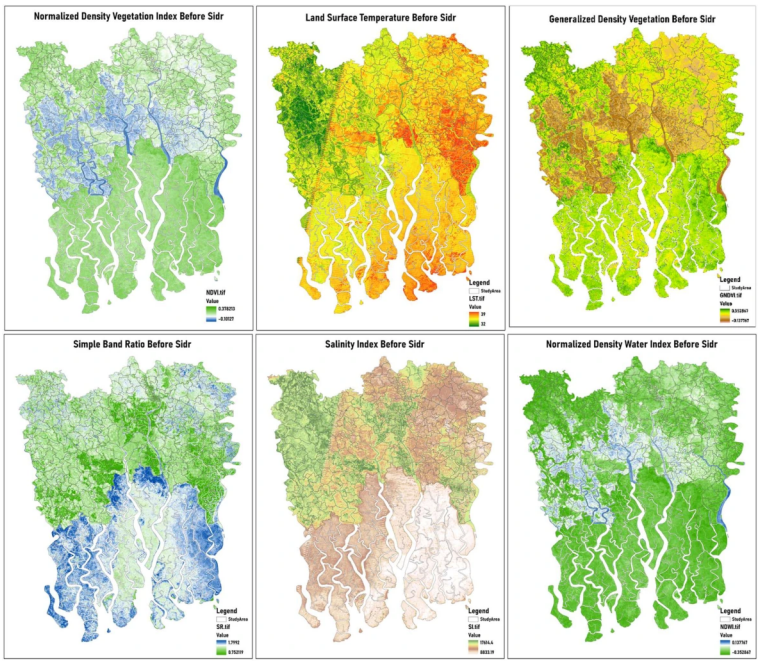
During my internship with UNDP's Sundarbans Area Based Programme, I developed a remote-sensing workflow to examine cyclone-related flooding across Khulna, Satkhira and Bagerhat using Landsat-derived environmental indicators and Random Forest regression.

## Study question

I tested whether satellite-derived environmental conditions could model cyclone-induced flood vulnerability across southwest coastal Bangladesh. Cyclone Sidr provided the main historical training context, and Cyclone Aila provided a second event for testing how well the relationship transferred.

## Inputs and model

I combined Landsat 5, 8 and 9 imagery with GNDVI, NDVI, NDWI, land-surface temperature, salinity index and simple-ratio indicators. I used temporal change in NDWI as a proxy for flood intensity and Random Forest regression to model nonlinear relationships between the environmental indicators and the flood signal.




Landsat-derived variables in the Cyclone Sidr context.

SIDR MODEL FIT

$R^2 = 0.83$

reported against NDWI-derived flood proxy

AILA MODEL FIT

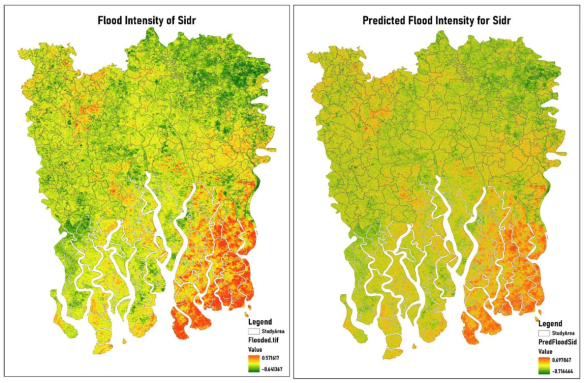
$R^2 = 0.79$

cross-event model evaluation

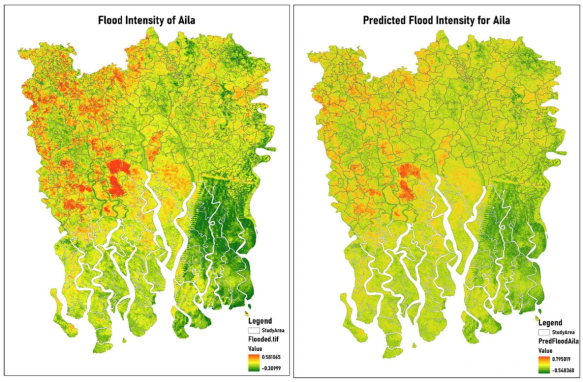
RESEARCH 02 · MODEL RESULTS

# Flood-vulnerability model results

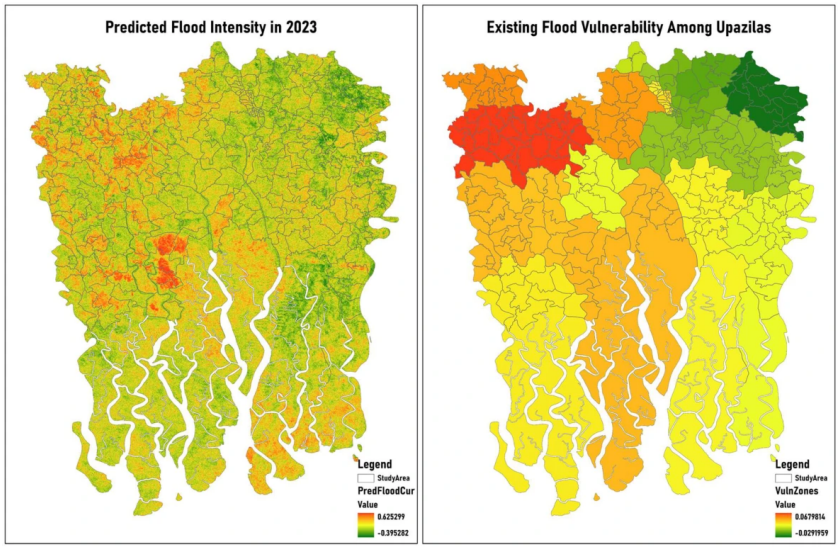
Cyclones Sidr and Aila: satellite-derived indicators and Random Forest predictions.




Sidr: satellite proxy versus model prediction.




Aila: satellite proxy versus model prediction.




Modelled 2023 flood-intensity proxy and upazila-level vulnerability output.

RESEARCH 03 · PUBLISHED PROCEEDINGS

# Spatial flood vulnerability of Greater Rangpur

I co-authored this GIS-based assessment of regional flood vulnerability in northern Bangladesh, combining socioeconomic and physical indicators for resilience planning.

EVIDENCE

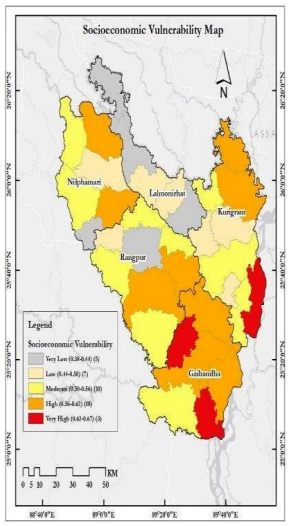
## 18 indicators

socioeconomic and physical indicators

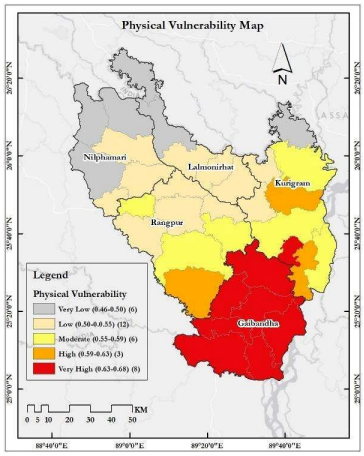
STUDY AREA

## Greater Rangpur

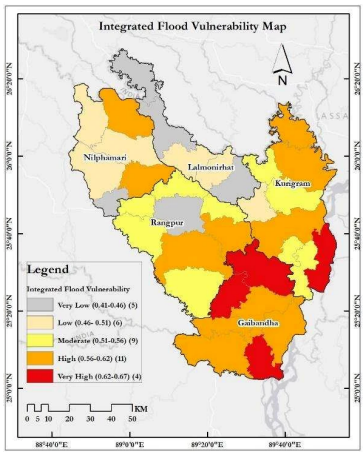
regional-scale vulnerability assessment




Socioeconomic component.




Physical component.




Integrated vulnerability map.

### Planning findings

We found clear spatial variation in vulnerability. Higher scores were associated with combinations of poverty, agricultural dependence, service and infrastructure gaps, low elevation and proximity to rivers. The index lets us compare where several disadvantage and exposure indicators overlap.

RESEARCH 04 · MANUSCRIPT UNDER REVIEW

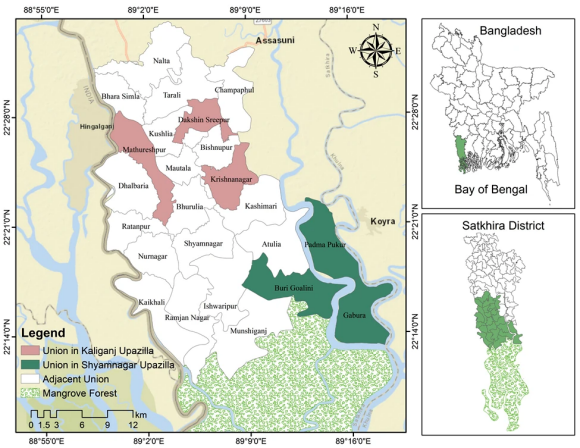
# Climate-related loss and damage in Satkhira

301 household responses across six unions, analysed with PCA-derived dimensions of loss and damage.

**SAMPLED HOUSEHOLDS**
301
survey responses

**STUDY AREAS**
6 unions
Shyamnagar and Kaliganj

**DIMENSIONS**
6
PCA-derived components




Six surveyed Satkhira unions and surrounding study area.

## Research question

How do coastal households experience loss and damage from climate-related hazards, and do the reported impacts differ among locations and sectors? We compared health, physical assets, productivity, social conditions, finances and water-related impacts. The study covers six unions in Shyamnagar and Kaliganj upazilas in Satkhira district.

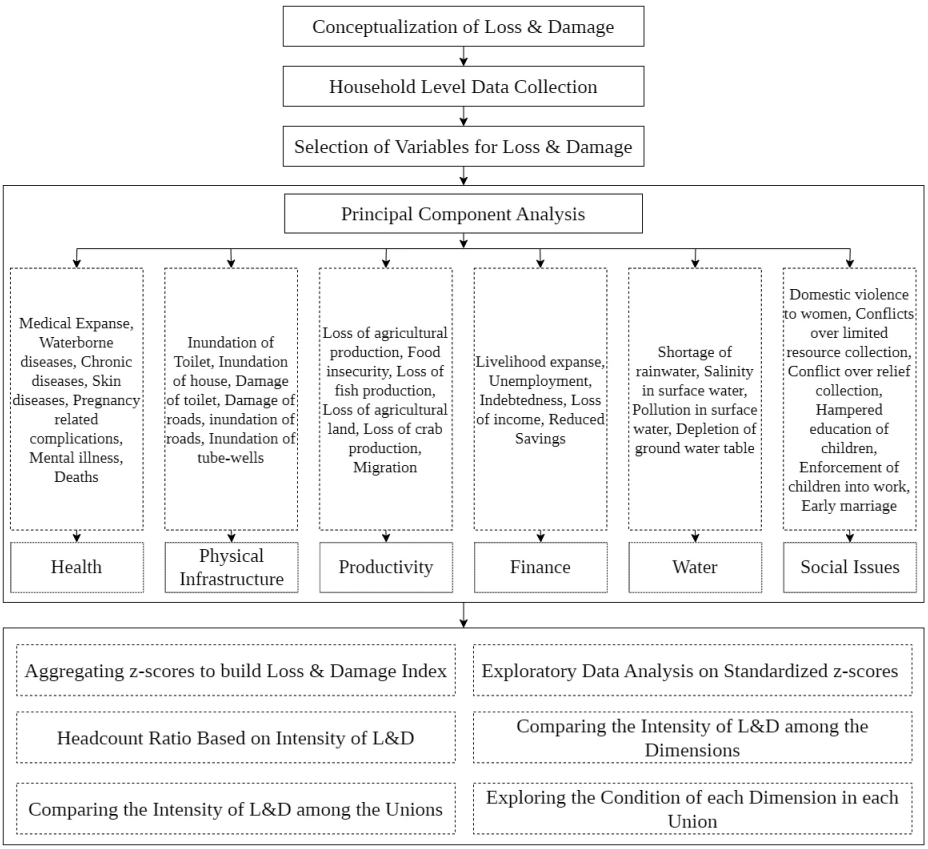
### Study geography

The sampled unions are Buri Goalini, Gabura, Padmapukur, Dakshin Sreepur, Krishnanagar and Mathureshpur in Satkhira district.

RESEARCH 04 · METHODOLOGY

# From household surveys to PCA indices

Survey indicators, principal component analysis and mapped loss-and-damage scores.




Household survey and PCA workflow.

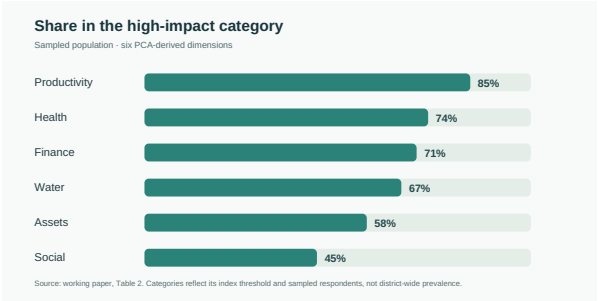
## Household indicators and PCA

The working manuscript describes selecting indicators from literature and expert input, reducing them using principal components analysis and Varimax rotation, then constructing a normalized household loss-and-damage index. The study also compares dimensions and unions using non-parametric tests, including Wilcoxon signed-rank and Mann–Whitney U tests.

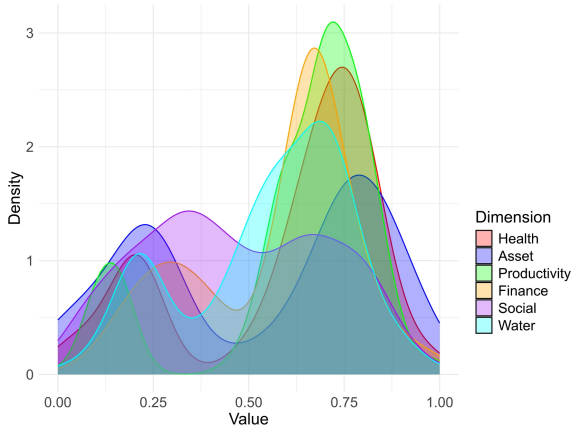
RESEARCH 04 · STATISTICAL RESULTS

# Six types of household impact

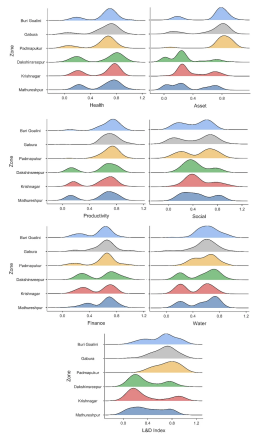
The six components include health, physical assets, productivity, social conditions, finances and water resources.




Share of surveyed households classified as highly impacted by dimension (working-draft Table 2).




Distribution of loss-and-damage scores across the six dimensions.



Distribution profiles by union and impact dimension.

## Main findings

Productivity-related losses had the largest share of the sampled population above the study's high-impact threshold. Health and financial impacts were also common. The draft reports higher asset and productivity loss-and-damage scores in the coastal unions of Buri Goalini, Gabura and Padmapukur. These comparisons describe survey responses, not population-wide prevalence or a causal estimate of climate change effects.

RESEARCH 04 · PCA RESULTS

# Principal component loadings

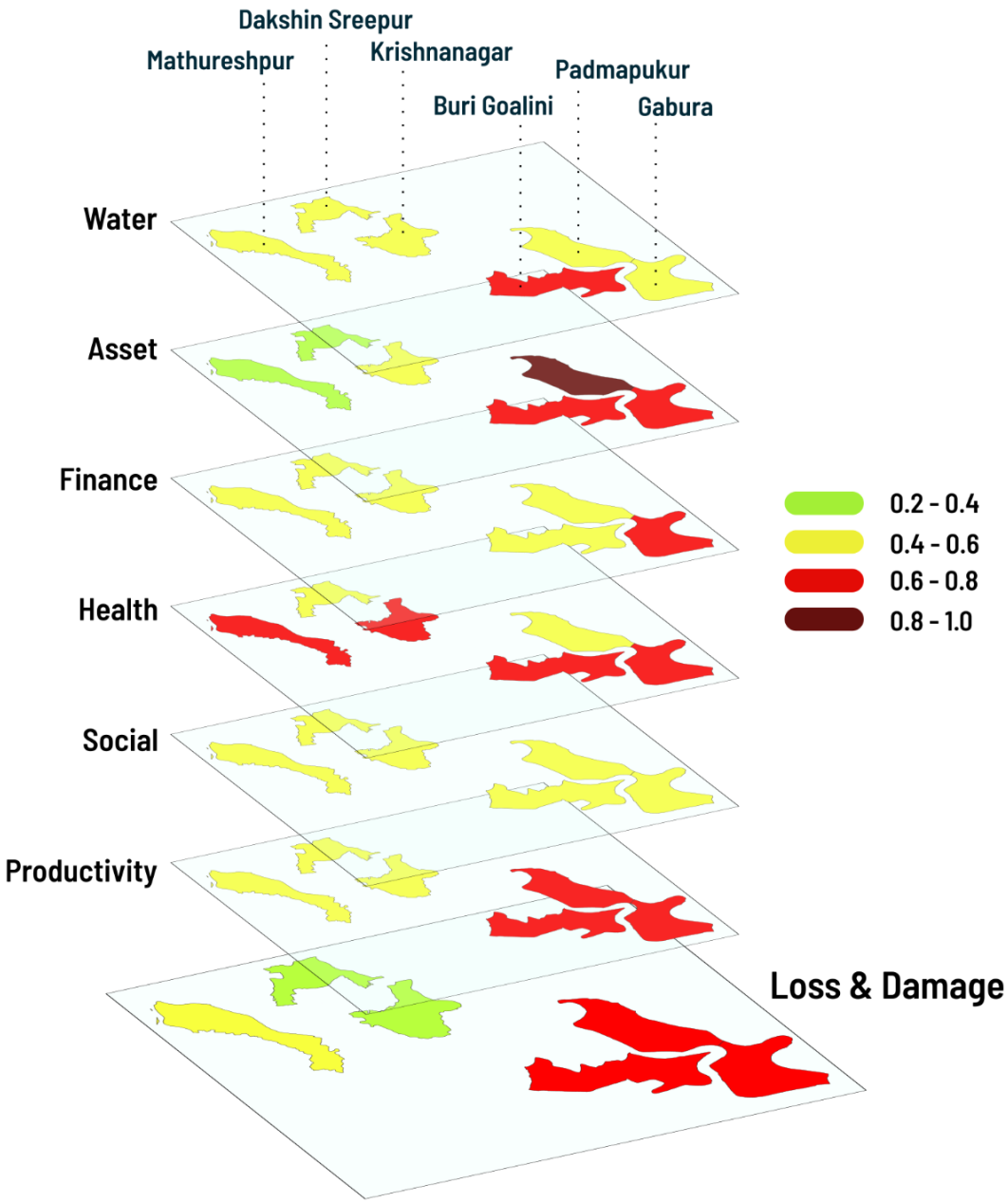
Indicators associated with the six components identified in the household survey.

| Variable / indicator | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| Increased medical expenses | 0.921 | | | | | |
| Increase in waterborne diseases | 0.893 | | | | | |
| Increase in chronic cardiac conditions | 0.847 | | | | | |
| Increase in skin diseases | 0.844 | | | | | |
| Pregnancy-related health concerns | 0.827 | | | | | |
| Increased mental illness | 0.743 | | | | | |
| Increase in deaths | 0.730 | | | | | |
| Inundation of toilets | | 0.942 | | | | |
| Inundation of houses | | 0.940 | | | | |
| Damage to toilets | | 0.938 | | | | |
| Damage to road culvert | | 0.935 | | | | |
| Inundation of road culverts | | 0.918 | | | | |
| Inundation of tube wells | | 0.885 | | | | |
| Loss of agricultural production | | | 0.861 | | | |
| Food insecurity | | | 0.858 | | | |
| Loss of shrimp production | | | 0.858 | | | |
| Loss of fish production | | | 0.853 | | | |
| Loss of agricultural land | | | 0.835 | | | |
| Loss of crab production | | | 0.810 | | | |
| Migration | | | 0.716 | | | |
| Increased domestic violence against women | | | | 0.845 | | |
| Increased conflicts over resource collection | | | | 0.844 | | |
| Increased conflicts over relief collection | | | | 0.842 | | |
| Disruption to children's education | | | | 0.816 | | |
| Children compelled to work | | | | 0.780 | | |
| Increase in early marriages | | | | 0.755 | | |
| Increased livelihood expenses | | | | | 0.875 | |
| Increased unemployment | | | | | 0.868 | |
| Indebtedness | | | | | 0.855 | |
| Loss of income | | | | | 0.833 | |
| Reduced savings | | | | | 0.789 | |
| Shortage of rainwater | | | | | | 0.839 |
| Increased salinity in surface-water sources | | | | | | 0.811 |
| Increased pollution of surface-water sources | | | | | | 0.780 |
| Depletion of the groundwater table | | | | | | 0.780 |

RESEARCH 04 · MAPPED RESULTS

# Spatial variation across six unions

Six loss-and-damage dimensions and a combined index are shown across the surveyed coastal unions.




Distribution of six loss-and-damage dimensions and the composite index across surveyed unions.

RESEARCH 05 · UNDERGRADUATE THESIS · 2023

# The Padma Bridge and urban growth

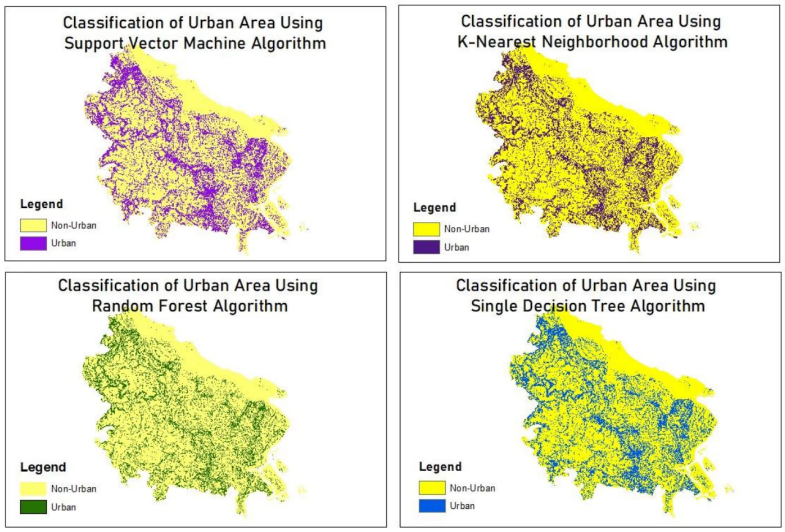
I examined how the Padma Bridge might affect the location and extent of future built-up land. With too little post-opening data to measure long-term change, I used the Jamuna Bridge's historical pattern as a modelling analogue.

## Research question

How might fixed road connectivity change the pattern of urban development south of the Padma Bridge? The study compared historical land-use change near the Jamuna Bridge before and after its opening, then adapted those patterns to the different geography and development conditions around the Padma Bridge. The Padma study area covered southern upazilas within roughly 50 km of the bridge.

## Land classification

I worked with four Landsat scenes from the Padma region (2010–2022) and six from the Jamuna region (1992–2022). I prepared training data to distinguish built-up and non-built-up land and compared Support Vector Machine, k-Nearest Neighbours, Random Forest and a single decision tree on the classification task. Random Forest performed best in the thesis comparison and was used for the subsequent land-cover series.




Classification model comparison for Padma-region imagery, original thesis Figure 3.7.

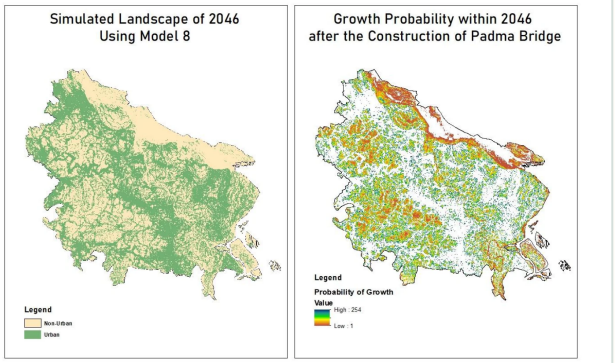
## Growth modelling

Using the classified land-cover maps, I estimated land-transition probabilities with a Markov model and calculated local development suitability through weights of evidence. I then ran Cellular Automata simulations in Dinamica EGO using patch and expander functions. The Jamuna-based model was adjusted for differences between the two bridge regions before being applied to the Padma scenarios.

RESEARCH 05 · MODELLED SCENARIOS

# Projected built-up growth to 2046

Modelled urban expansion under the thesis scenario, 2022–2046.




2046 simulated landscape and growth probability (thesis Figure 4.15).




Upazila-level predicted additional growth (thesis Figure 4.17).

SCENARIO RESULT

## 317 km²

conditional additional built-up area

MODEL FAMILY

## Markov + CA

with weights of evidence and spatial predictors

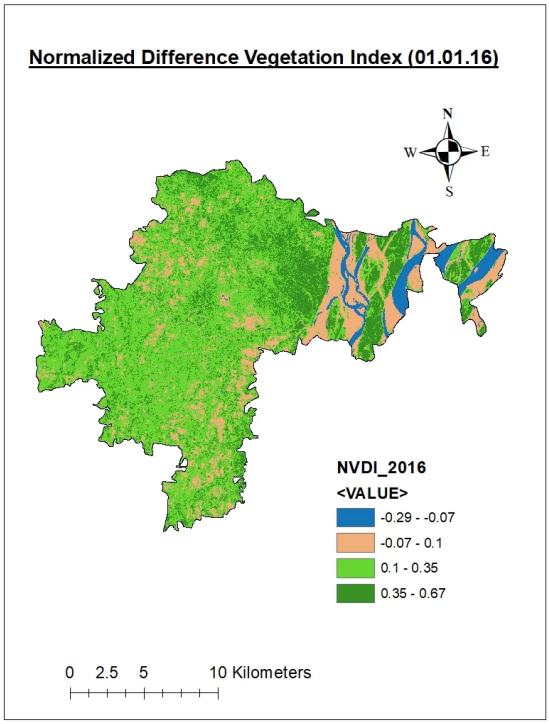
### Results

The alternative scenarios indicated uneven pressure on future development. Gopalganj Sadar, Madaripur Sadar and Shibchar were among the areas with the strongest projected response. These are simulated outcomes, not built-up land observed after the bridge opened.

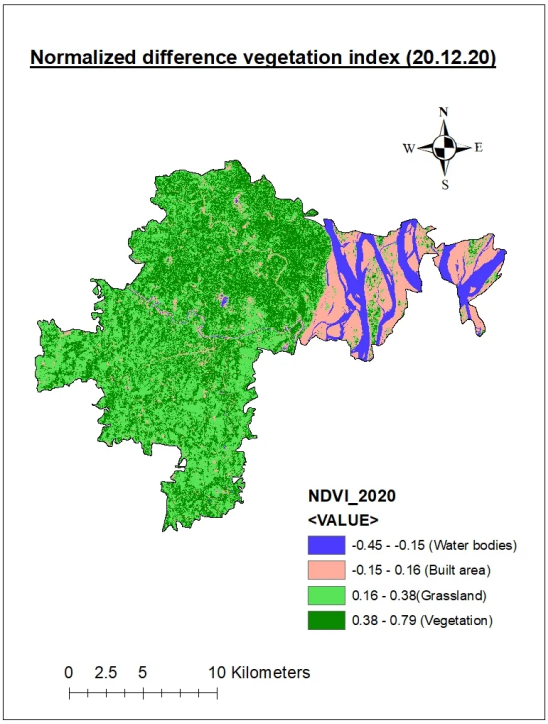
APPLIED 01 · REMOTE-SENSING COURSEWORK

# Gaibandha land-cover change

In this coursework project, I used multi-date satellite imagery, land-cover classification, NDVI and change detection to examine land-cover conditions between 2016 and 2020.




Normalized Difference Vegetation Index, 2016.




Normalized Difference Vegetation Index, 2020.

## Workflow

I produced supervised and unsupervised classification outputs, class-area calculations and vegetation-index maps. The exercise helped me compare how different remote-sensing products describe spatial change.

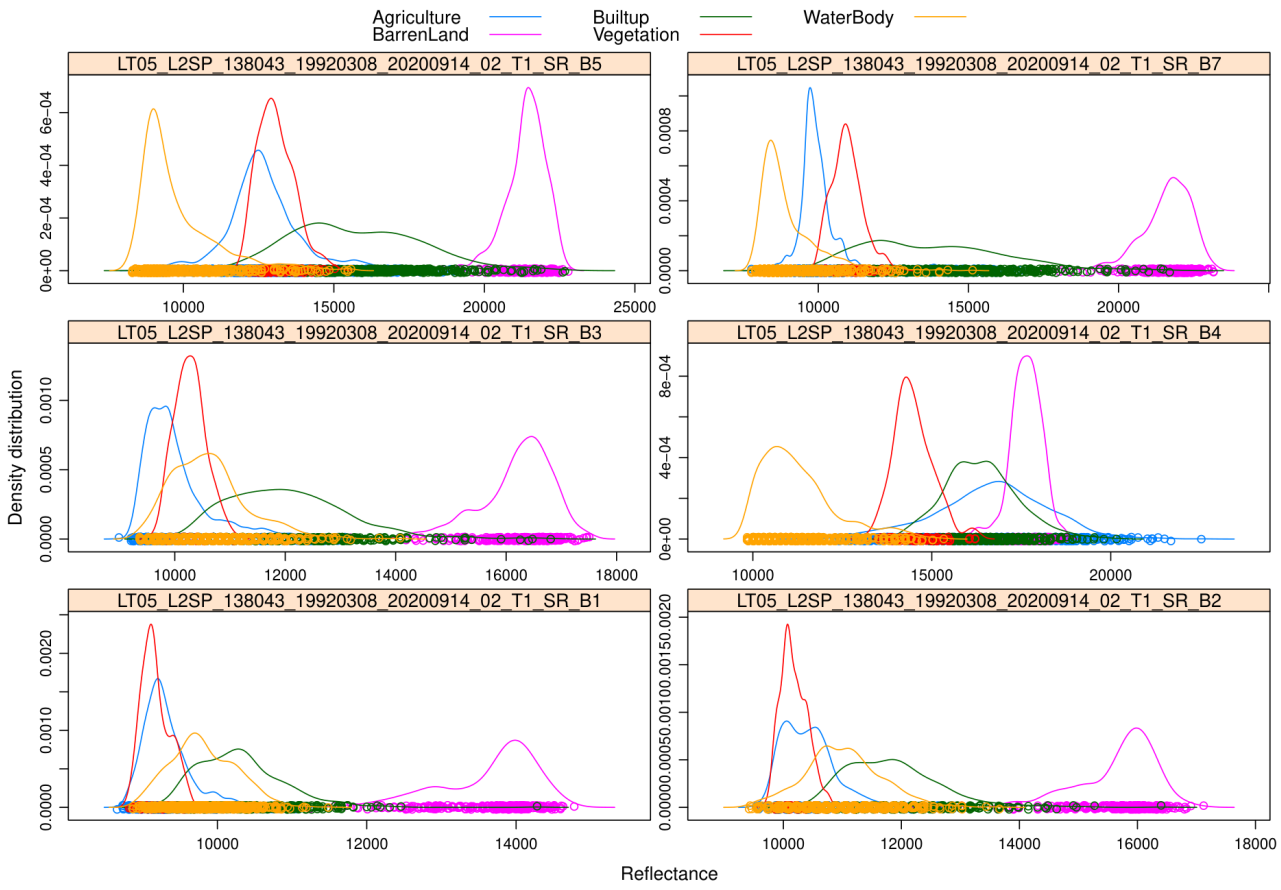
## Study question

I examined how major land-cover classes changed between 2016 and 2020, using vegetation indices and classification outputs to interpret the change.

APPLIED 01 · SPECTRAL ANALYSIS

# Landsat 5 spectral distributions

Landsat 5 scene acquired 8 March 1992, showing spectral variation across five land-cover classes.

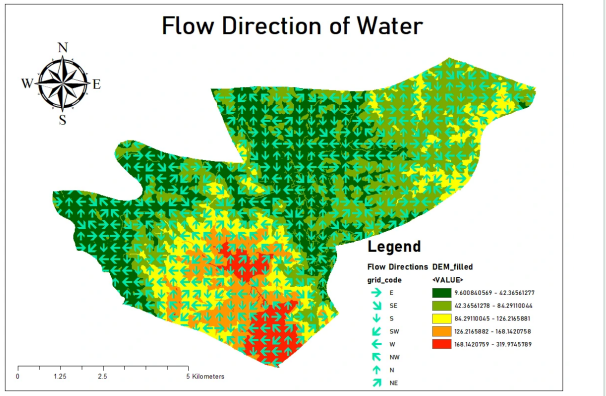



Classwise Landsat 5 B1–B5 and B7 distributions for agriculture, barren land, built-up land, vegetation and water body classes.

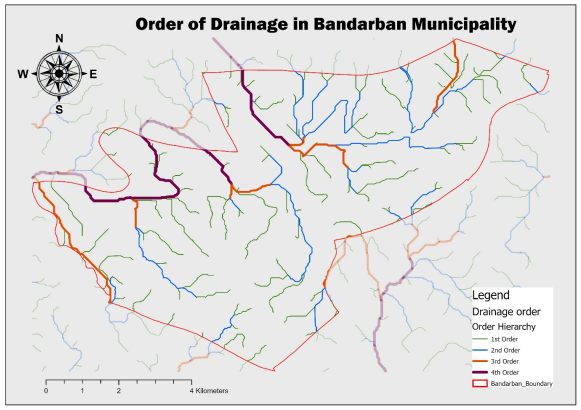
APPLIED 02 · GIS HYDROLOGY

# Terrain-based drainage mapping

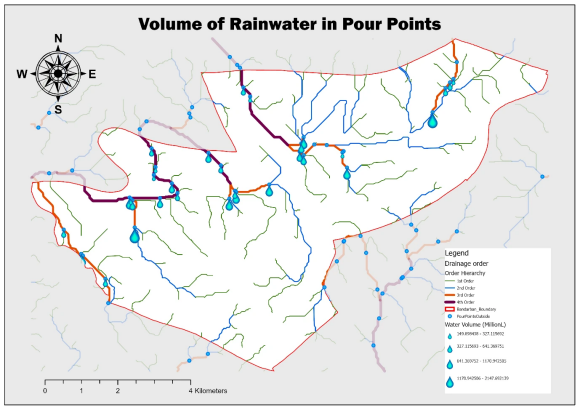
Bandarban Municipality · natural drainage, flow direction, stream hierarchy and catchments.



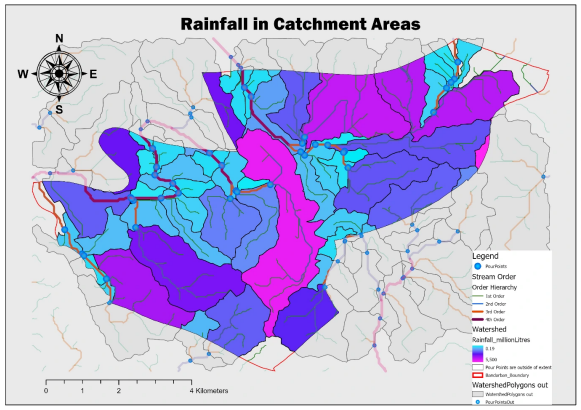

Flow direction model.




Extracted stream hierarchy.




Pour points and outlets.




Delineated catchments.

## Planning question

Urban drainage follows terrain rather than administrative boundaries. I used elevation to identify likely surface-flow paths, stream hierarchy, outlets and contributing catchments.

APPLIED 03 · TRANSPORT COURSEWORK

# Sukundirbag Tower traffic assessment

For this individual traffic-impact coursework project, I assessed a proposed office development in Gazipur using traffic counts, passenger-car-unit conversion, trip generation, background growth and site-access recommendations.




Morning traffic movement diagram.




Evening traffic movement diagram.

## Traffic survey & PCU conversion

I recorded cycles, motorcycles, vans, rickshaws, CNGs, cars, buses and trucks, then converted the mixed traffic to passenger-car units so I could compare directional flows on a common basis.

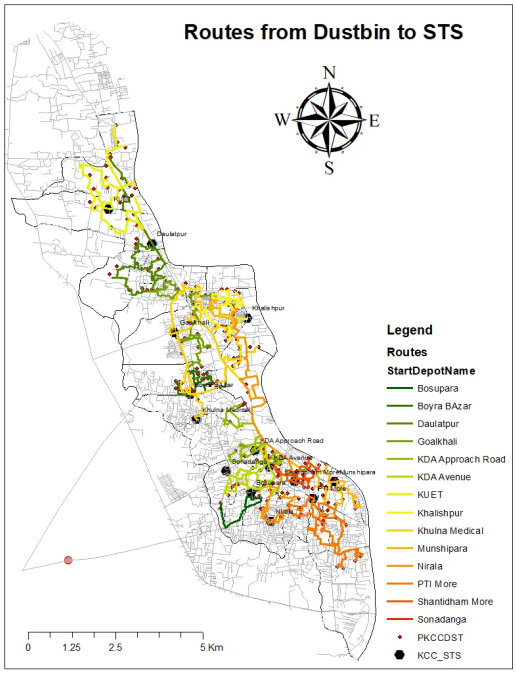
## Future traffic scenario

I used a five-year background-growth assumption and compared future directional demand with the capacity assumptions in the report. My recommendations focused on access management, markings and junction operation.

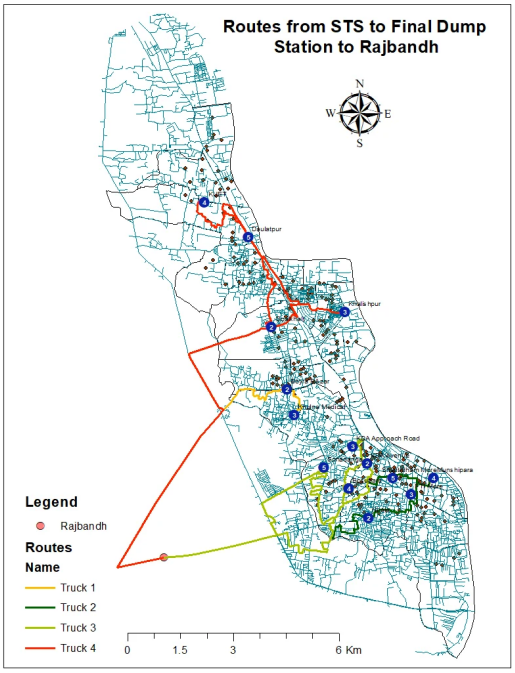
APPLIED 04 · GIS NETWORK ANALYSIS

# Municipal solid-waste routing

In this municipal-services project, I examined the spatial relationship between waste-generating points, transfer locations, vehicle movement and the final disposal route.




Routing to secondary transfer stations.




Routing to final disposal location.

## Collection system

I represented a three-stage municipal collection system linking waste-generating points to primary collection or dustbin locations, secondary transfer stations and a final treatment or disposal location.

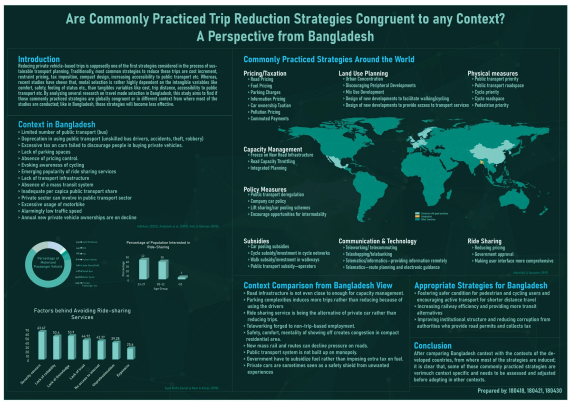
## Routing analysis

I used GIS network analysis to examine how collection vehicles could move between transfer locations and the final disposal site. This connected service planning with road-network constraints and the spatial distribution of collection points.

APPLIED 05 · POLICY SYNTHESIS

# Trip-reduction strategies

I reviewed common travel-demand strategies in a Bangladesh context, focusing on the local conditions that affect whether those strategies are likely to work as intended.




Original transport policy synthesis poster.

## Research question

I asked how well common demand-management measures transfer to Bangladesh when public-transport quality, road safety, enforcement, travel behaviour and institutional conditions differ from the settings in which many of those policies were first evaluated.

## Main argument

My main conclusion was that trip-reduction strategies need to match the local mechanisms that shape travel behaviour. Pricing, compact development, ride sharing and active travel can have different effects depending on transit reliability, safety, access and trip purpose. I therefore focused on context instead of assuming that a policy label transfers unchanged between cities.

ACADEMIC BACKGROUND

# Research outputs and experience

Selected publications, research appointments and applied project experience.

## PUBLICATION STATUS

### CONFERENCE PROCEEDINGS

**Assessing Spatial Flood Vulnerability of Greater Rangpur Region in Bangladesh Using GIS: A Tool for Resilience Planning at Regional Scale**

Himel, T. I., Talukder, R. H., Hossain, M. Z., Naz, S., Arfin, S., & Fariha, J. N. In Planning for Inclusiveness and Sustainability in Post Pandemic Era . Bangladesh Institute of Planners. ISBN 978-984-35-2613-7.

### UNDER REVIEW

**Integrating GIS and Multi-Criteria Decision Analysis to Assess Climate Change and Disaster-Induced Loss and Damage in Coastal Bangladesh**

Talukder, R. H., Rahman, M. A., Arfin, S., Fariha, J. N., Husna, A. U., Ha-mim, N. M., Hossain, M. Z., & Rahaman, K. R. Environmental Development (Elsevier; manuscript under review).

## RESEARCH EXPERIENCE

**Research Assistant (part-time)**
Jun 2021 – Dec 2024

**University Grants Commission of Bangladesh**
Applied GIS, land-cover analysis and historical-spatial research to campus development, urban land-use change, environmental conditions and the physical evolution of Dhaka.

**Intern, Sundarbans Area Based Programme**
Feb 2023 – Jul 2023

**United Nations Development Programme (UNDP) · Khulna**
Designed a remote-sensing and machine-learning workflow to predict cyclone-induced flood vulnerability in Khulna, Satkhira and Bagerhat using historical cyclone scenarios and satellite-derived environmental indicators.

**Qualitative Researcher**
Aug 2023 – Oct 2023

**Tiller Bangladesh · Dhaka**
Designed and conducted interviews, focus-group discussions and key-informant interviews on fishing practices, socioeconomic conditions and conservation knowledge.

**Technical Consultant**
Mar 2021 – Jun 2021

**Centre for Environmental and Geographic Information Services (CEGIS) · Dhaka**
Digitized census-related spatial datasets and developed GIS layers for administrative boundaries, settlements, infrastructure and socioeconomic indicators.

ACADEMIC BACKGROUND

# Methods, teaching and contact

Training in urban and rural planning, applied GIS and mixed-methods research.

EDUCATION

## Bachelor of Urban and Rural Planning

Khulna University · completed 2023

THESIS

Predicting the Influence of the Padma Bridge on Future Urban Growth.

## METHODS AND SOFTWARE

| | |
|---|---|
| **GIS & Earth observation** | ArcGIS Pro, QGIS, ArcPy, Google Earth Engine, ERDAS IMAGINE, Dinamica EGO; spatial analysis; land-use/land-cover analysis; remote-sensing indices; raster/vector processing; cartographic communication. |
| **Geospatial machine learning, programming & data analysis** | Python, R, SPSS, Jamovi, NVivo; Random Forest regression; quantitative analysis; qualitative coding; data visualization; model evaluation using $R^2$, MAE and RMSE. |
| **Transportation & spatial modelling** | Counterfactual analysis, induced-demand research, scenario-based forecasting, uncertainty analysis, network and accessibility concepts; TransCAD, MATSim and VISSIM exposure through research and teaching. |
| **Research methods** | Research design; literature review; quantitative and qualitative methods; interviews; focus groups; key-informant interviews; thematic analysis; spatial equity analysis; research philosophy; research ethics; academic writing and citation management. |

CONTACT

## Rifat Hossain Talukder

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rifatht.github.io